

Robust AI-based intrusion detection for IoMT: A data-efficient approach via optimized feature selection and hybrid data balancing

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ABSTRACT

The growing integration of the Internet of Medical Things (IoMT) in critical infrastructure demands efficient and accurate Intrusion Detection Systems (IDS) tailored to their resource-constrained environments. In this work, we propose the Optimized Preprocessing Framework (OPF): A novel pre-processing methodology to develop a computationally optimized and robust against extreme class imbalance Artificial Intelligence-based IDS. Our methodology operates in two stages: First, a fully balanced dataset is created using Random Undersampling of the majority classes, and a Gini feature-based prioritization method is employed to efficiently reduce the feature space. Then, a Random Forest model is trained on a nearly balanced version of the dataset, generated using a hybrid resampling technique via logarithmic interpolation that employs SMOTE-based oversampling of minority classes and Random Undersampling of majority classes. This process preserves relative class order while compressing extreme disparities. Moreover, it significantly reduces the feature space and computational overhead. We validate the proposed approach on multiple real-world datasets, including CICIoMT2024, IoMT-TrafficData, and CICIoT2023. Results show that our method improves the F1-score metric, while reducing model complexity by 31.11%, 31.39%, and 78.16% in each dataset respectively, thereby making it suitable for deployment in practical IoMT environments.

1. Introduction

The integration of digital technologies in the healthcare sector has established a new ecosystem: The Internet of Medical Things (IoMT). This transformative ecosystem includes interconnected medical devices, sensors, and software applications [1]. This new paradigm enables unprecedented capabilities, including continuous remote patient monitoring, automated collection and analysis of physiological data, and the enhancement of diagnostic and therapeutic procedures [2]. Furthermore, the adoption of IoMT has been rapid and widespread, with estimates suggesting that approximately 60% of healthcare organizations have already integrated IoMT devices into their operations, resulting in over 300,000 interconnected medical devices, from smart insulin pumps to connected pacemakers [3].

However, while IoMT promises to enhance patient care and operational efficiency, it simultaneously creates an extensive and complex attack surface [4]. The critical nature of these devices and the transfer of sensitive patient data between systems make IoMT a high-value target for malicious actors. The spectrum of cyber threats is broad, the most

reported attacks encompass Denial-of-Service (DoS) and Distributed Denial-of-Service (DDoS) that can disable life-sustaining equipment; spoofing, and data injection attacks, intercepting sensitive data transmissions; and exploitation, unauthorized access, and password attacks that exploit weaknesses in system vulnerabilities and authentication mechanisms, posing serious risks to both patient data and overall operational integrity [5]. In fact, a successful attack on an IoMT network can have severe, life-threatening consequences, not only to data privacy but also posing significant risks to patient safety [6].

In response to these threats, Artificial Intelligence (AI)-based Intrusion Detection Systems (IDS) have emerged as a promising defense mechanism. Unlike traditional IDS, which can be ineffective against novel cybersecurity attacks, AI-based IDS employs Machine Learning (ML) and Deep Learning (DL) to model normal network behavior and anomalous deviations, detecting a broad number of sophisticated and zero-day attacks, which makes this approach indispensable for securing dynamic IoMT environments [7,8].

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Despite their potential, the practical deployment of AI-based IDS in IoMT is challenging, which raises a fundamental conflict between the resource-intensive nature of advanced AI models and the inherent limitations of IoMT environments. On the one hand, IoMT devices, ranging from wearable sensors to embedded medical devices, are typically designed with constraints on computational power (CPU), memory capacity, and energy consumption (battery life). These limitations make the deployment of complex models impractical at the network edge, where fast pre-processing and real-time detection are most needed [9]. On the other hand, the network traffic data generated within IoMT environments presents a considerable amount of network characteristics, many of which may be redundant or irrelevant to the task of intrusion detection. Feature selection techniques should be applied to reduce the feature space to only relevant network characteristics. Moreover, the real nature of data within IoMT environments constitutes an extreme imbalance among classes, including benign traffic and different cyberattacks [10]. This reality creates a pressing demand for security solutions that are not only effective but also robust and computationally efficient in reducing the sensitivity against class imbalance.

The current literature contains numerous studies proposing AI-based IDS for IoT and IoMT environments. These studies often address the challenges of high dimensionality and class imbalance, typically by employing feature selection and data balancing techniques, respectively [11,12]. However, these pre-processing techniques, when they are not treated as isolated, but as interconnected components of a unified strategy, can generate optimized models [13]. Therefore, there is a clear need to develop an integrated pre-processing framework that combines feature selection and data balancing. Such a framework should aim to produce a robust and data-efficient model that is well-distributed, enabling an unbiased and accurate classifier.

This paper introduces the Optimized Pre-processing Framework (OPF), a novel approach designed specifically to fill this research gap. The OPF is a two-stage framework that holistically addresses the dual challenges of IoMT data. The first stage employs a Gini feature-based prioritization method to efficiently reduce the feature space, retaining only the most important features. The second stage introduces a hybrid resampling technique that utilizes SMOTE-based oversampling of minority classes and random undersampling of majority classes via logarithmic interpolation. This methodology ensures that resampling is non-linear but smooth, preserving relative class order while compressing extreme disparities. Thus, the primary contributions of this paper are:

1. The proposal of a novel, two-stage pre-processing framework that integrates Gini feature-based selection and data balancing to optimize data representation for AI-based IDS in IoMT environments.
2. The introduction of a hybrid resampling strategy that employs an empirical grid search calibration to adjust the number of samples in a given dataset, thereby avoiding overfitting while preserving the original class distribution.
3. A demonstration of the framework's ability to facilitate a robust and data-efficient IDS that is robust to biased class distributions by significantly reducing feature dimensionality without sacrificing model performance.
4. A validation of the proposed model on three recent publicly available IoMT/IoT datasets (i.e., CICIOT2023, CICIOMT2024, and IoMT-TrafficData), including a comparison against multiple baseline models.

The remainder of this paper is structured as follows: Section 2 provides a review of the background and related work in the field. Section 3 presents a description of the proposed OPF. Section 4 outlines the experiment design, including datasets, evaluation metrics, and comparative models. Section 5 presents and analyzes the experimental results. Finally, Section 6 concludes the paper and discusses potential directions for future research.

The main notations used in this work are summarized in Table 1.

Table 1

List of notations.

Notation	Definition
D	The given dataset.
C	The total number of classes in the dataset.
c	A specific class within the dataset.
p_c	The probability of samples to be linked into class c at a given node.
$Gini(D)$	The Gini impurity of the dataset D .
N_{target}	The reference value (anchor) for the resampling thresholds, computed using the geometric mean of the class sizes.
N_c	The number of samples of a specific class c .
T_c	The new target size for class c mapped using logarithmic interpolation.
T_{min}	The minimum threshold set to avoid aggressive oversampling and overfitting of minority classes.
T_{max}	The maximum threshold set to compress the influence of majority classes safely.
N_{min}	The class count of the smallest class in the dataset.
N_{max}	The class count of the largest class in the dataset.
K	The ranked features set obtained after Gini importance extraction ($K = \{k_1, k_2, \dots, k_n\}$).
k	The specific number of features evaluated or selected.
M_0	The preliminary Random Forest model trained on the initial balanced dataset.
M^*	The final optimized and robust IDS model.
X	The actual value of a numerical feature before scaling.
X_{min}	The smallest value in the dataset for a specific feature.
X_{max}	The largest value in the dataset for a specific feature.
X_{scaled}	The scaled value of the feature after applying Min-Max normalization.

2. Related work

In this section, we review the recent literature surrounding machine learning-based intrusion detection in IoT and IoMT networks. To establish the foundational context and identify the critical gaps addressed by our OPF, we explicitly rely on the comprehensive analysis conducted in our recent systematic literature review [5], which extensively evaluated behavior-based intrusion detection systems and their operational constraints in IoMT environments.

Machine learning has become a cornerstone in the development of IDS for IoT and IoMT environments. Numerous studies propose ML and DL-based IDS as a means of detecting sophisticated and evolving threats that traditional signature-based solutions cannot address. For example, a SaaS-based IDS framework for healthcare IoT employs feature engineering and optimization methods to adapt detection to real-world healthcare data [11]. Similarly, decision-based trees and recurrent neural architectures have been introduced for cyber-attack detection in IoT traffic, demonstrating their ability to model temporal dependencies and outperform classical approaches [14,15]. Other works propose intrusion detection tailored to remote healthcare applications [12] or scalable frameworks such as SmartEdge, which leverages ML for IoMT anomaly detection at the edge [16]. Deep learning has also been explored, with convolutional neural networks (CNNs) applied to IoMT traffic [17] and hybrid architectures combining autoencoders with ML classifiers for anomaly detection [18]. Recent studies have also explored highly complex architectures; for instance, robust frameworks leveraging hybrid CNN-LSTM models integrated with Deep Reinforcement Learning have achieved exceptional binary classification accuracy on datasets like CICIOMT2024 [19]. Finally, the

use of federated learning with privacy-preserving techniques is also explored in IoMT environments [20]. These approaches collectively show that ML-driven IDS can enhance detection accuracy, yet they also reveal important methodological limitations when confronted with the realities of IoT and IoMT datasets.

Pre-processing plays a central role in ensuring IDS models remain efficient and effective under constrained conditions. Many works investigate feature engineering and dimensionality reduction as mechanisms to control complexity. Particle Swarm Optimization (PSO) has been applied both for feature subset selection and for hyperparameter tuning in IDS pipelines [11,15]. Walrus Optimization Algorithm (WaOA) has also been explored, offering superior results, particularly in hyperparameter optimization and improving model performance [21]. Other approaches combine Principal Component Analysis (PCA) with heuristic methods such as Grey Wolf Optimization (GWO) to derive compact feature representations suitable for deep neural models [22]. Similarly, ‘Top-K’ feature selection approaches have demonstrated that tree-based models like Random Forest and Extreme Gradient Boosting can maintain an F1-score of over 93% on the CICIoMT2024 dataset while drastically reducing processing time by focusing only on the 10 most discriminating characteristics [23]. SmartEdge integrates multiple selection techniques, including PCA, Random Forest importance, and Gradient Boosting, to optimize feature input for IoMT anomaly detection [16]. However, a significant portion of the literature either omits explicit feature selection [17,18] or applies it in a limited manner, resulting in unnecessarily large feature spaces. In one study, anomaly detection models for IoMT referenced feature selection but without detailing the techniques applied [24], while others relied primarily on CNNs or autoencoders to implicitly extract features from raw data [17, 18]. This highlights a recurring issue: although feature selection is recognized as beneficial, its systematic integration into IDS design remains inconsistent.

Random Forest (RF) models are particularly notable in this context due to their dual role as both classifiers and feature ranking tools. RF has been employed in IoMT anomaly detection pipelines where feature importance scores guide the reduction of input dimensionality [24]. SmartEdge also incorporates RF-based importance in conjunction with PCA and boosting to refine the most relevant traffic attributes [16]. RF’s ability to rank features through Gini index scoring offers a transparent and computationally efficient method for prioritizing network attributes. Yet, in most studies, this capability is used only as a supplementary step rather than a central component of the pre-processing pipeline, limiting its potential impact on improving both robustness and efficiency.

Another critical dimension of IDS research in IoT and IoMT lies in addressing class imbalance. Real-world datasets are heavily skewed, with attacks such as DoS or DDoS dominating, while stealthy threats like spoofing or data injection remain underrepresented. To mitigate this, some works experiment with synthetic balancing techniques. For instance, Wasserstein GAN with Gradient Penalty (WGAN-GP) has been used to generate synthetic minority attack samples, thereby augmenting limited real-world data [12]. SMOTE has also been applied to IoMT datasets, showing improved detection of underrepresented attacks [24]. Other hybrid methods like SMOTEENN and SMOTETomek are increasingly employed to clean the dataset after oversampling, removing ambiguous or noisy samples that cross overlapping class boundaries [25]. Despite these contributions, many studies either neglect balancing altogether or only mention it briefly. In botnet detection studies, no explicit balancing is applied, risking bias toward frequent attack classes [26]. Similarly, hybrid models based on CNNs or autoencoders are trained directly on imbalanced data without correction [17, 18]. This inconsistency undermines the generalizability of results, as models trained on skewed distributions tend to underperform when deployed against diverse, real-world environments.

Taken together, the surveyed literature highlights both progress and gaps in ML-based IDS research for IoT and IoMT. There is a

broad consensus that ML can significantly improve detection accuracy and adaptability, but the approaches are fragmented. Feature selection, while often mentioned, is inconsistently applied, with few works leveraging RF importance systematically despite its advantages. Balancing techniques are similarly inconsistent, with most studies either oversampling or undersampling but rarely exploring hybrid or adaptive methods that could better preserve class structure. Furthermore, explainability remains almost entirely absent: despite the complexity of modern ML models, none of the surveyed works employed methods such as SHAP (Shapley Additive Explanations) to interpret feature contributions. These gaps point to the need for an integrated framework that jointly addresses feature space reduction and class imbalance in a systematic manner, while remaining lightweight enough for constrained IoT and IoMT deployments.

3. The proposed optimized pre-processing framework (OPF)

The core contribution of this research lies in the creation of the Optimized Pre-processing Framework (OPF), a systematic, two-stage approach that combines feature selection and dataset balancing techniques, overall creating a robust and data-efficient IDS that reduces the sensitivity against class imbalance. This design aims to overcome the high-dimensional and imbalanced IoMT network traffic data into a compact and well-distributed representation, suitable for environments that require fast pre-processing and real-time detection. The main purpose of the framework is to first simplify the problem space through intelligent feature reduction, and then to rebalance this simplified space in a data-driven manner. An overview of the diagram illustrating the end-to-end pipeline of the system is presented in Fig. 1.

3.1. Framework overview

The OPF pipeline consists of the following phases:

1. **Data Collection:** The process begins with the collection of raw network traffic from IoMT datasets, which are characterized by high dimensionality and class imbalance. The datasets used to validate our framework are detailed in Section 4.
2. **Data Cleaning:** This second step consists of cleaning the raw data and preparing it according to our needs. In particular, the number of classes, the type of data encoding, and the data normalization we want to use are chosen. In Section 4, we describe these parameters in more detail.
3. **Data Pre-Processing:** The pre-processing of data is a crucial step before training and validating the ML model; the data collected and cleaned must be correctly prepared to feed the model with accurate information. Therefore, the OPF is implemented in this step, and it can be divided into two major stages.

- **Stage 1: Gini Feature-based Prioritization.** This initial stage aims to identify and select a subset of the most important features. It leverages the embedded feature importance mechanism of a Random Forest classifier trained on a fully balanced dataset using Random Undersampling, thereby reducing dimensionality, computational cost for the subsequent stages, and mitigating the bias of the feature selection process towards majority classes. The Random Forest classifier was used for its consistent high performance while being robust and simple in IoMT environments [10,27,28]. This stage is described in more detail in Section 3.2.
- **Stage 2: Hybrid Resampling Strategy.** This second stage addresses the class imbalance of datasets. It employs a novel hybrid resampling strategy that combines SMOTE-based oversampling with Random Undersampling to create a nearly balanced dataset. The final sampling ratios are not predetermined, but found through an empirical calibration process that maximizes model performance. This stage is detailed in Section 3.3.

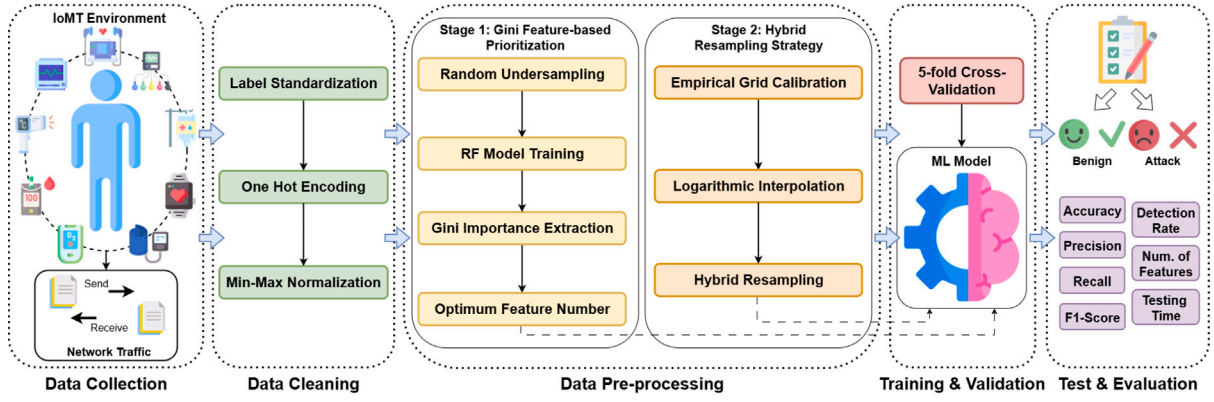


Fig. 1. Overall architecture of the proposed system.

4. Training and Validation: With the output of the OPF, we produce an optimized dataset for training and validating the ML model. Section 4.3 describes the methodology employed in this phase in more detail.
5. Test and Evaluation: After proper validation, the model with higher performance metrics and detection rates is finally evaluated using the test set, leading to more reliable generalization metrics (see Section 5 for more detail).

A key aspect of this design is the relationship between the two stages of the OPF. The initial feature reduction in Stage 1 is a critical step that enhances the effectiveness of Stage 2. By performing feature selection first, the subsequent application of hybrid resampling is more reliable and efficient to generate a robust, low-dimensional AI-based IDS that is able to reduce the sensitivity against class imbalance. In the next subsections, the logic behind these stages is explained in detail.

3.2. Stage 1: Gini feature-based prioritization

The objective of this stage is to identify and get the optimum feature subset from the original high-dimensional dataset. This is achieved through the following steps:

1. Initial Dataset Balancing: To train an unbiased model for feature selection, a fully balanced dataset is first created. The distinct classes within the dataset are identified using the standardized target variable labels (established during the data cleaning phase) before any resampling. Therefore, the dataset balancing is accomplished using simple Random Undersampling, where all majority classes are undersampled to match the number of instances in the smallest minority class. This method is chosen for its computational simplicity and its unbiased manner of feature selection towards the majority classes [29].
2. Preliminary RF Model Training: A standard Random Forest classifier is trained on this temporary, balanced dataset. The RF is chosen for its robustness and ability to effectively and efficiently compute feature importance using the Gini index [30].
3. Gini Importance Extraction: The main part of this stage is to extract the feature importance scores from the trained RF model. The importance of each feature is quantified using the Gini index. First, the Gini index measures the impurity of a specific dataset subset (or node) D containing C classes. If p_c is the probability of a sample belonging to class c at a given node, the Gini impurity of the dataset would be described as follows in Eq. (1). To calculate the specific importance of each individual feature, the RF algorithm evaluates the decrease in this Gini impurity every time that feature is selected to split a node. The total decrease in Gini impurity achieved by a specific feature is aggregated across all nodes where it was used, and then

averaged across all the decision trees in the forest. Therefore, a feature that consistently produces pure node splits will result in a larger aggregated decrease in Gini impurity, granting it a higher feature importance score. This metric effectively quantifies how well a feature can distinguish between distinct classes in the dataset [31,32].

$$Gini(D) = 1 - \sum_{c=1}^C p_c^2 \quad (1)$$

4. Feature Selection: The Gini importance scores are used to create a list of all features, ordered from most to least important. This ranked list forms the basis for our feature selection strategy: During model validation, we will systematically assess performance using incrementally larger subsets of these top-ranked features (e.g., the top 10, top 15, top 20, and so on). The objective is to identify the optimal number of features that maximizes model performance while minimizing model complexity.

3.3. Stage 2: Hybrid resampling strategy

After simplifying the feature space, Stage 2 focuses on improving the class imbalance of the dataset using a data-driven methodology.

This stage focuses on a hybrid resampling methodology, employing SMOTE Oversampling [33] for minority classes to increase representation, and Random Undersampling [34] for majority classes to prevent biased detection and overfitting towards majority classes. Current datasets present highly imbalanced class distributions; training directly on original distributions can bias classifiers and suppress underrepresented behaviors. To mitigate this, we introduce a log-scale resampling strategy via geometric mean that emphasizes fairness and efficiency in the training process. This procedure also helps to retain more of the original data distribution's characteristics, preserving relative class order while compressing extreme disparities.

3.3.1. Defining resampling boundaries (T_{min} and T_{max})

Before calculating the new class distributions, we must first establish the boundaries for our resampling strategy to prevent over-manipulation of the data. For each dataset, we define a minimum threshold (T_{min}) and a maximum threshold (T_{max}) that respect the following theoretical constraints:

- $T_{min} \leq 10 \times N_{min}$ to avoid aggressive oversampling and overfitting of minority classes.
- $T_{max} \in [5\%, 15\%]$ of N_{max} to allow strong yet safe majority class compression.

The selected lower and upper thresholds aim to strike a principled balance: enhancing minority class learnability without incurring

overfitting, and mitigating majority class dominance while preserving representative diversity. These decisions are grounded in prior studies on the limitations of aggressive oversampling [35–37], and the effectiveness of clustering-based or hybrid undersampling strategies [38–40]. With this, we also aim to identify robust configurations that maintain high classification stability across minority and majority classes while optimizing model performance, minimizing bias, and resource cost.

3.3.2. Anchor estimation via geometric mean (N_{target})

With the boundaries established, we compute a reference value (N_{target}) for the resampling thresholds (T_{min} , T_{max}) that acts as a secure anchor for the resampling process. N_{target} should theoretically lie within the previously defined $[T_{min}, T_{max}]$ range. To calculate this anchor, we compute the geometric mean [41] of the class sizes, as presented in Eq. (2):

$$N_{target} = \exp\left(\frac{1}{|C|} \sum_{c=1}^{|C|} \ln N_c\right) \quad (2)$$

This metric reflects the multiplicative structure of imbalanced data, being a good approach for data imbalance problems [42]. It serves as a logarithmic midpoint of the dataset's class distribution, which acts as a balanced anchor to smoothly scale the sampling ratios. Moreover, it favors a more conservative, targeted sampling that improves minority detection while limiting harm to majority performance [43].

3.3.3. Class-wise balancing via logarithmic interpolation (T_c)

To address severe class imbalance while preserving the intrinsic statistical structure of network traffic data, we adopt a resampling strategy that adjusts class sizes using a logarithmic interpolation approach. Previous studies show that class occurrences in cybersecurity and IoT traffic datasets typically follow heavy-tailed distributions, where disparities between classes are multiplicative rather than additive [28,44,45]. Consequently, linear rescaling of sample counts may not reflect the real distribution of classes. For that reason, class sizes are first mapped into logarithmic space, which reduces these extreme disparities, leading to a more balanced and stable training process. Therefore, using the defined boundaries and the geometric anchor, each class count N_c is mapped to a new target size T_c using the following interpolation formula (Eq. (3)):

$$T_c = T_{min} + (T_{max} - T_{min}) \times \frac{\ln N_c - \ln N_{min}}{\ln N_{max} - \ln N_{min}} \quad (3)$$

Where N_{min} and N_{max} are the class counts of the smallest and largest classes, respectively. This interpolation ensures a non-linear but smooth transition that preserves the relative class order while drastically compressing extreme disparities.

3.3.4. Pseudo-code of the proposed OPF

The overall pseudo-code of the OPF is presented in Algorithm 1, where the different steps of the entire pipeline, from data collection, cleaning, and pre-processing to training, validation, and final testing of the model, are observed.

4. Experimental setup

To evaluate the performance and efficiency of the proposed OPF, a comprehensive experimental setup was designed. This section details the datasets used for training, validation, and testing, the evaluation metrics, and the specific evaluation methodology details required for reproducibility.

Algorithm 1 Pipeline of the Proposed System

Require: IoMT dataset D

Ensure: Optimized Random Forest model M^*

- 1: **Step 1: Data Collection and Cleaning**
 - Collect raw network traffic data from an IoMT environment.
 - Standardize labels, encode variables, normalize features.
- 2: **Step 2: Stage 1 – Feature Ranking**
- 3: Balance D using **Random Undersampling** $\rightarrow D_{balanced}$.
- 4: Train Random Forest M_0 on $D_{balanced}$.
- 5: Extract feature importance scores via **Gini Index**.
- 6: Obtain ranked feature set $K = \{k_1, k_2, \dots, k_n\}$.
- 7: **Step 3: Stage 2 – Hybrid Resampling & Training**
- 8: **for each** $(T_{min}, T_{max}) \in \mathcal{G}$ **do** ▷ Thresholds
- 9: **for each** $k \in \{k_1, k_2, \dots, k_{max}\}$ **do** ▷ Features set
- 10: Generate $D_{nearly}(k, T_{min}, T_{max})$ by:
 - Pick samples via Logarithmic interpolation
 - SMOTE oversampling of minority classes
 - Random undersampling of majority classes
- 11: Train $M(k, T_{min}, T_{max})$ on $D_{nearly}[F_{1..k}]$.
- 12: Validate $M(k, T_{min}, T_{max})$ on validation set.
- 13: Store performance metrics $\mathcal{P}(k, T_{min}, T_{max})$.
- 14: **end for**
- 15: **end for**
- 16: **Step 4: Model Validation**
- 17: Select optimal configuration $(k^*, T_{min}^*, T_{max}^*)$ that maximizes performance.
- 18: Define final model $M^* = M(k^*, T_{min}^*, T_{max}^*)$.
- 19: **Step 5: Final Test & Evaluation**
- 20: Test M^* on unseen test data.
- 21: Report:
 - Performance metrics: Accuracy, Precision, Recall, F1-score, Detection Rate via confusion matrix.
 - Efficiency metrics: #Features, Testing time.
- 22: **Output:** Optimized and robust IDS M^* .

4.1. Datasets

The analysis presented in this work uses three publicly available datasets: CICIOT2023 [45], CICIOT2024 and [28], created by the University of New Brunswick (Canada) in 2023 and 2024, respectively, and the IoMT-TrafficData [44], developed by the Polytechnic of Liria (Portugal) in 2024.

The CICIOT2023 is intended as a real-time benchmark dataset for large-scale attacks in an IoT environment and was generated by means of an IoT testbed including 105 devices of different brands and types (e.g. cameras, smart plugs, hubs, sensors). The dataset includes normal (benign) traffic and 33 different attack types that are grouped into 7 major attack categories, as depicted in Table 2.

The CICIOT2024 dataset extends the scope of IoT intrusion detection benchmarks with a particular emphasis on the Internet of Medical Things (IoMT) domain. The dataset was generated using a heterogeneous testbed of around 40 real and simulated IoT/IoMT devices, including smart healthcare devices such as patient monitors, infusion pumps, and IoT gateways, connected through diverse protocols like Wi-Fi, MQTT, and Bluetooth. This setup enables the dataset to realistically capture the unique security challenges faced in medical IoT environments. CICIOT2024 contains both benign traffic and a narrower set with 18 types of attacks grouped into 5 categories, but includes protocol-specific threats such as MQTT-based attacks (see Table 2). The inclusion of MQTT-based attacks is highly relevant, as the MQTT protocol is heavily used in IoMT environments for telemetry data (e.g., patient monitors).

The IoMT-TrafficData dataset, as the CICIOT2024, is focused on network traffic generated by medical IoT devices and wearable sensors. However, it considers not only IP-based communications but also

Table 2
Attacks included in CICIoT2023, CICIoMT2024 and IoMT-TrafficData datasets.

Attack type	Attack subtype	CICIoT2023	CICIoMT2024	IoMT-TrafficData
DDoS	TCP, UDP, SYN, ICMP flood	45.34%	66.63%	✗
	ACK, UDP and ICMP fragmentation	2.19%	✗	✗
	RSTFIN, PSHACK flood	17.44%	✗	✗
	SynonymousIP flood	7.71%	✗	✗
	HTTP flood	0.06%	✗	✗
	SlowLoris	0.05%	✗	2.02%
	SlowRead, ApacheKiller, R-U-Dead-YET	✗	✗	11.99%
		72.79%	66.63%	14.01%
DoS	TCP, UDP, SYN, ICMP flood	17.33%	25.32%	✗
Reconnaissance	OS, Port and Vulnerability scan	0.47%	1.49%	(not specified)
	Ping sweep	0.005%	0.011%	(not specified)
	Host discovery	0.29%	✗	(not specified)
		0.76%	1.5%	17.61%
Web-based	SQL, Command injection	0.02%	✗	✗
	Backdoor malware, Uploading attack	0.01%	✗	✗
	Cross-site scripting, Browser hijacking	0.02%	✗	✗
		0.05%		
CAM Table Overflow		✗	✗	12.84%
Spoofing	ARP spoofing	0.66%	0.2%	4.1%
	DNS spoofing	0.38%	✗	✗
		1.04%	0.2%	4.1%
Brute force	Dictionary	0.03%	✗	✗
Mirai	UDPPplain, GREIP and Greeth flood	5.64%	✗	✗
MQTT-based	DoS connect and Publish flood	✗	0.78%	(not specified)
	Malformed data	✗	0.08%	✗
	DDoS connect and Publish flood	✗	2.86%	(not specified)
			3.72%	35.2%

Bluetooth, reflecting the heterogeneous connectivity commonly found in healthcare environments. The dataset consists of benign traffic along with eight distinct attack types. While it contains fewer attack types, it uniquely features threats such as Bluetooth reconnaissance and injection, as well as CAM (Content Addressable Memory) table overflow, which can disrupt local hospital switching networks containing medical devices. Another key distinction is that the IoMT-Traffic Data dataset contains data from the CoAP protocol.

Table 2 presents the types of attacks considered in each dataset. For each attack, the table specifies whether it is included in a given dataset and, if so, indicates the percentage of samples it represents with respect to the entire dataset, along with the proportion of benign traffic. Values highlighted in bold indicate the percentage of samples corresponding to each attack category. The IoMT-Traffic Data dataset also includes a BLE traffic subset containing two types of attacks. However, these samples are stored in a separate dataset and have been evaluated independently from the IP-based dataset. Because of that, we do not include these samples in the table.

In addition to the types of devices and attacks considered, an essential distinction among these datasets lies in the features or attributes they include. The CICIoT2023 and CICIoMT2024 datasets contain 47 and 45 attributes, respectively. In contrast, the IoMT-Traffic data considers a total of 64 attributes.

4.2. Evaluation metrics

The evaluation metrics used for assessing the proposed models are described in this section. Several metrics have been proposed for comparing not only performance but also efficiency of proposed models for the specific characteristics of IoMT environments.

4.2.1. Performance metrics

Several evaluation metrics have been proposed in the literature for assessing AI-based intrusion detection systems (IDS) in attack detection tasks. These metrics, which are commonly used in classification problems, are derived from the fundamental measures of True Positives

(TP), True Negatives (TN), False Positives (FP), and False Negatives (FN) [46]. The metrics used in this study are as follows: Accuracy for measuring the proportion of correctly predicted instances out of the total number of instances. Precision, which is used for quantifying the proportion of correct positive predictions. Recall is used to measure the proportion of actual positive instances that the model correctly identifies. And F1-score, which is the harmonic mean of precision and recall, and provides a single metric that balances both metrics and can be computed as shown in Eq. (4).

$$F1\text{-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

When one or a few classes heavily outweigh others, accuracy can be misleading since this metric is dominated by the majority classes, thus masking the failure of the model in classifying minority classes. In such cases, the F1-score serves as a more reliable metric because it evaluates how well the model identifies minority classes by considering both the correctness of positive predictions (precision) and the ability to capture actual positives (recall). Furthermore, in multiclass classification, a confusion matrix is an $N \times N$ table (where N is the number of classes) that summarizes the performance of a model by comparing its predicted labels against the true labels for all classes. Each row of the matrix represents the actual class, while each column represents the predicted class.

4.2.2. Efficiency metrics

Performance evaluation of the models in terms of efficiency was based on two main metrics: test time and the number of features (k). Test time measures how efficiently an algorithm can make predictions on unseen data, providing insight into its computational performance during deployment. The number of features, in turn, indicates the dimensionality and complexity of the data used by the model. Considering both metrics allows for assessing algorithms that not only perform efficiently but also rely on a manageable set of input variables, balancing accuracy and practicality.

Table 3Balancing thresholds (T_{min} , T_{max}) tested for each dataset according to N_{min} , N_{max} and N_{target} .

Dataset	N_{min}	N_{max}	N_{target}	T_{min}	T_{max}
CICIoMT2024	14,233	4,677,298	200k	[75k, 100k, 125k]	[250k, 500k, 700k]
IoMT-TrafficData	7211	1,312,031	90k	[35k, 50k, 65k]	[100k, 150k, 200k]
CICIoT2023	2090	5,437,530	108k	[10k, 15k, 20k]	[250k, 550k, 800k]

4.3. Evaluation methodology

We adopted a standard train-validation-test split methodology across all datasets to ensure rigorous and unbiased experiments. Each dataset was initially split into the training/validation sets (80%) and the test set (20%). This split was performed using stratified and shuffling sampling to ensure that the original class distribution was maintained in each set. The roles of these sets were:

- Training/Validation set (80%): This set was used exclusively for all model development tasks. This includes hyperparameter tuning of our OPF framework via a 5-fold cross-validation process.
- Test set (20%): This set remained unseen and untouched during all training and tuning phases. It was used only once for the final performance evaluation of the optimized models to provide an unbiased evaluation of their ability to generalize to new, previously unseen data.

4.4. Implementation details

To ensure reproducibility and clarity of our experimental results, this section details the data preparation pipeline, the model selection, and the specific software and hardware environment used in this study. It is important to note that all data pre-processing steps (such as min-max normalization, SMOTE, and Random Undersampling) were applied strictly after the train-validation-test split to guarantee no data leakage for the model.

4.4.1. Data preparation and dataset handling

As presented in Fig. 1, a multi-step data cleaning process was applied to each dataset before the data pre-processing stage. These are:

1. Label Standardization: First, the class labels were standardized to create a consistent multiclass classification scheme specific to each dataset. Mid-level attack classes were selected in all datasets to provide actionable, meaningful detection categories without fragmenting the data into sub-classes that have too few samples to train a robust model.
2. One Hot Encoding: The target classes, which were initially categorical features, were transformed into a numerical format using One-Hot encoding. This prevents the model from assuming any ordinal relationship between these categories.
3. Min-Max Normalization: Finally, all numerical features were scaled to a uniform range of [0, 1] using Min-Max Normalization. The equation of the Min-Max scaling is presented in Eq. (5), where X is the actual value, and X_{min} and X_{max} are the smallest and largest values in the dataset for that specific feature, respectively. This step is crucial to ensure that features with larger value ranges do not over-influence the model's learning process.

$$X_{scaled} = \frac{X - X_{min}}{X_{max} - X_{min}} \quad (5)$$

In addition to this multi-step data cleaning process, specific handling was required for two of the datasets due to their unique characteristics and scale:

- CICIoT2023: Due to the substantial size of this dataset and the resulting computational constraints, a representative 20% master subset was created. To ensure this subset accurately reflected the original data's characteristics, it was generated via stratified random sampling, which meticulously preserves the class distribution of the full dataset. All experiments for CICIoT2023 were conducted on this common, shuffled subset.
- IoMT-TrafficData: This dataset provides traffic data in two formats: IP-based packets and IP-based flows. For this study, we utilized only the IP-based flows dataset, as it provides a suitable feature space for our experiment.

4.4.2. Model selection and hyperparameters

The Random Forest (RF) classifier was selected as the core algorithm for all experiments. This choice is supported by its consistent high performance, robustness, and simplicity in AI-based IDS for IoMT environments [10,27]. Our own preliminary experiments also validated this, showing that RF demonstrates a superior balance of performance and efficiency to other models such as Decision Trees (DT), eXtreme Gradient Boosting (XGB), and Feed Forward Neural Networks (FNN). Moreover, the hyperparameters used for the RF model were: $\{criterion='gini', n_estimators=100\}$. Nevertheless, it is important to highlight that the proposed OPF is considered a model-agnostic methodology that can be applied to any other supervised ML model.

The data balancing boundaries for our OPF framework (i.e., T_{min} and T_{max}) were key hyperparameters. The search space for these boundaries was defined empirically for each dataset to reflect its unique class distributions. The specific groups of thresholds validated are detailed in Table 3, which represent the specific sets of experimental parameters that were validated during our grid search. For both T_{min} and T_{max} , we specifically defined groups of three experimental values to test a spectrum of intervention intensities while fulfilling the theoretical constraints stated in Section 3.3.

On the one hand, the T_{min} threshold groups in each dataset were chosen to avoid aggressive oversampling and overfitting. For that reason, the first value on the group was selected to be conservative ($\approx 5 \times N_{min}$), the second value to have a moderate oversampling ($\approx 7 \times N_{min}$), and the third one to have the maximum allowed oversampling ($\approx 10 \times N_{min}$). On the other hand, the T_{max} threshold groups in each dataset were chosen to allow strong yet safe majority class compression. For the first boundary, we wanted to have maximum compression ($\approx 5\%$ of N_{max}); for the second boundary, we wanted a balanced reduction in class samples ($\approx 7.5\%$ of N_{max}); finally, the third boundary was selected to have minimal compression ($\approx 15\%$ of N_{max}).

4.4.3. Experimental environment

All experiments were conducted on a server with the following specifications to ensure consistency:

- Hardware: The server used was an Intel Xeon Processor, with 32 CPUs and 126 GB of RAM.
- Software: The implementation was developed in Python (v3.10.12). The data manipulation and ML tasks were performed using Scikit-learn (v1.6.1), Pandas (v2.2.3), and NumPy (v2.2.5). Visualizations were generated using Matplotlib (v3.10.3).

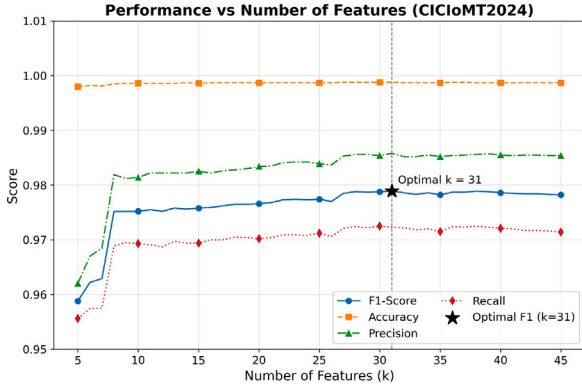


Fig. 2. Performance of the model in CICIoMT2024 dataset for each specific number of features, according to optimum balancing threshold $T_{min} = 75$ k, $T_{max} = 700$ k.

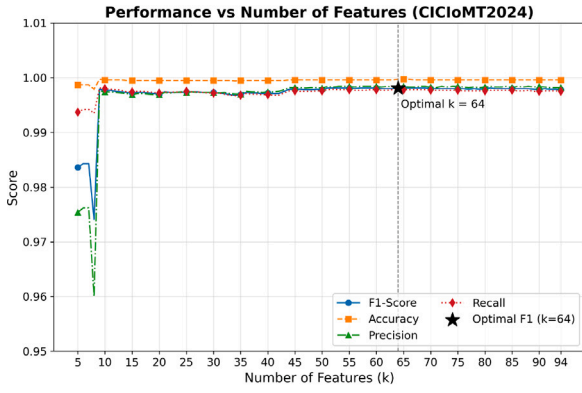


Fig. 3. Performance of the model in IoMT-TrafficData dataset for each specific number of features, according to optimum balancing threshold $T_{min} = 50$ k, $T_{max} = 200$ k.

5. Results and discussion

The results of the OPF are presented in this section and are divided into two main parts. Firstly, we explain the model optimization results that led us to define the specific thresholds for the hybrid resampling and the optimal number of features in each dataset. Then, a performance evaluation and comparative analysis are conducted, comparing the OPF with established baseline models and state-of-the-art models. The limitations and challenges faced during the development and evaluation of the framework are also presented in this section.

5.1. Model optimization

The primary evaluation took place during the validation phase, where we needed to determine the optimal data-specific thresholds for the hybrid resampling and the optimal number of features associated with it. Several experiments were made with the CICIoMT2024, IoMT-TrafficData, and CICIoT2023 datasets, according to the 9 different threshold combinations presented in Section 4. In particular, features are added iteratively in descending order of Gini importance, and the optimal subset is chosen where the F1-score is optimum before degrading due to noise inclusion. The overall tuning process to get the final model in each dataset with the best results from the 9 threshold combinations is presented in Tables 4, 5, and 6.

Firstly, for the CICIoMT2024 dataset, $T_{min} = 75$ k and $T_{max} = 700$ k with $k = 31$ features was the distribution that gave overall best results in the validation test, being the best model in terms of accuracy,

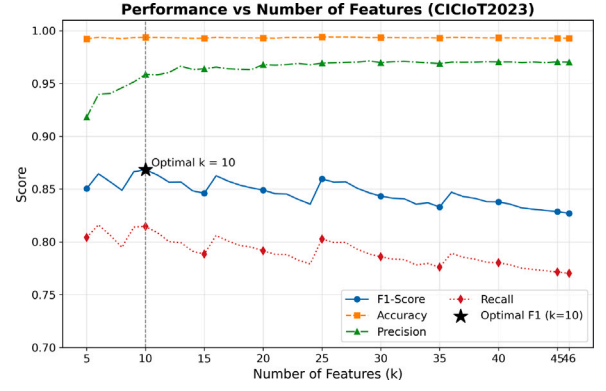


Fig. 4. Performance of the model in CICIoT2023 dataset for each specific number of features, according to optimum balancing threshold $T_{min} = 20$ k, $T_{max} = 550$ k.

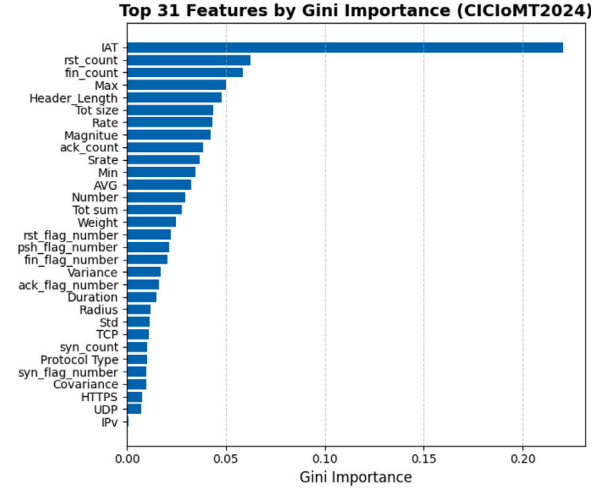


Fig. 5. Top 31 features by Gini importance used in the CICIoMT2024 dataset.

precision, and F1-score. Secondly, for the IoMT-TrafficData dataset, the model with the best overall performance was for the $T_{min} = 50$ k and $T_{max} = 200$ k with $k = 64$ features. This model outperformed the others regarding accuracy, recall, and F1-score metrics. Finally, in the CICIoT2023 dataset, the model with $k = 10$ features, $T_{min} = 20$ k, and $T_{max} = 550$ k demonstrated higher performance than the others, especially in terms of recall and F1-score.

Figs. 2, 3, and 4 illustrate the overall performance of the model among the different set of features, with the best threshold combination for CICIoMT2024 ($T_{min} = 75$ k and $T_{max} = 700$ k), IoMT-TrafficData ($T_{min} = 50$ k and $T_{max} = 200$ k), and CICIoT2023 ($T_{min} = 20$ k and $T_{max} = 550$ k) datasets, respectively. The optimal number of features where model performance peaked is also highlighted in the aforementioned figures. It can be observed how the performance of the models changes when training and evaluating them with a different set of features. In particular, a similar behavior is noticed in all three figures: In terms of the accuracy score, it is almost constant among all features. However, the precision, recall, and F1-score performance metrics increase during a specific number of features, until reaching their maximum score; afterward, the score decreases until reaching the maximum number of features in each dataset. As particularly noted in Fig. 4 (CICIoT2023 dataset), the inclusion of additional features beyond the optimal point does not cause a smooth decline, but rather an irregular decrease, specifically in the recall and F1-score metrics. This irregular performance drop occurs because the incremental addition of these lower-ranked features introduces specific noise and redundant

Table 4

Validation results according to the tuning process for threshold selection in CICIoMT2024 dataset.

Min. threshold	Max. threshold	Optimal features	Accuracy	Precision	Recall	F1-score
$T_{min} = 75k$	$T_{max} = 250k$	$k = 36$	0.9985	0.9809	0.9750	0.9779
	$T_{max} = 500k$	$k = 28$	0.9987	0.9847	0.9734	0.9789
	$T_{max} = 700k$	$k = 31$	0.9988	0.9858	0.9723	0.9789
$T_{min} = 100k$	$T_{max} = 250k$	$k = 36$	0.9984	0.9797	0.9753	0.9775
	$T_{max} = 500k$	$k = 37$	0.9987	0.9830	0.9740	0.9785
	$T_{max} = 700k$	$k = 36$	0.9988	0.9843	0.9734	0.9787
$T_{min} = 125k$	$T_{max} = 250k$	$k = 36$	0.9984	0.9789	0.9754	0.9771
	$T_{max} = 500k$	$k = 29$	0.9986	0.9821	0.9743	0.9782
	$T_{max} = 700k$	$k = 38$	0.9987	0.9833	0.9736	0.9784

Table 5

Validation results according to the tuning process for threshold selection in IoMT-TrafficData dataset.

Min. threshold	Max. threshold	Optimal features	Accuracy	Precision	Recall	F1-score
$T_{min} = 35k$	$T_{max} = 100k$	$k = 54$	0.9996	0.9980	0.9979	0.9979
	$T_{max} = 150k$	$k = 56$	0.9996	0.9984	0.9978	0.9981
	$T_{max} = 200k$	$k = 57$	0.9996	0.9985	0.9978	0.9981
$T_{min} = 50k$	$T_{max} = 100k$	$k = 53$	0.9996	0.9980	0.9979	0.9979
	$T_{max} = 150k$	$k = 51$	0.9996	0.9983	0.9979	0.9981
	$T_{max} = 200k$	$k = 64$	0.9997	0.9984	0.9979	0.9981
$T_{min} = 65k$	$T_{max} = 100k$	$k = 53$	0.9996	0.9981	0.9979	0.9980
	$T_{max} = 150k$	$k = 53$	0.9996	0.9983	0.9979	0.9981
	$T_{max} = 200k$	$k = 64$	0.9996	0.9985	0.9978	0.9981

Table 6

Validation results according to the tuning process for threshold selection in CICIoT2023 dataset.

Min. threshold	Max. threshold	Optimal features	Accuracy	Precision	Recall	F1-score
$T_{min} = 10k$	$T_{max} = 250k$	$k = 10$	0.9931	0.9612	0.8128	0.8680
	$T_{max} = 550k$	$k = 10$	0.9938	0.9633	0.8130	0.8678
	$T_{max} = 800k$	$k = 10$	0.9940	0.9621	0.8133	0.8674
$T_{min} = 15k$	$T_{max} = 250k$	$k = 10$	0.9930	0.9564	0.8138	0.8679
	$T_{max} = 550k$	$k = 10$	0.9937	0.9595	0.8134	0.8674
	$T_{max} = 800k$	$k = 10$	0.9939	0.9602	0.8133	0.8671
$T_{min} = 20k$	$T_{max} = 250k$	$k = 10$	0.9928	0.9535	0.8142	0.8677
	$T_{max} = 550k$	$k = 10$	0.9936	0.9587	0.8146	0.8684
	$T_{max} = 800k$	$k = 10$	0.9939	0.9590	0.8143	0.8678

data that disproportionately degrades the model's ability to accurately identify highly imbalanced minority classes. Consequently, while overall accuracy and precision might remain relatively stable, the model's recall, and thus the harmonic F1-score, fluctuates and drops as the feature space becomes larger. This clearly indicates that the presented OPF approach is able to identify irrelevant and noisy features that actively harm minority class detection. Furthermore, by training and evaluating the model with a smaller feature subset, we reduce feature dimensionality while improving overall model performance. This is achieved with $k = 31$, $k = 64$, and $k = 10$ features in the CICIoMT2024, IoMT-TrafficData, and CICIoT2023 datasets, respectively. The list of the top features used in each dataset is detailed in Figs. 5, 6, and 7.

It is important to note that selecting a non-optimal number of features degrades performance: retaining too many features reintroduces intra-class noise that harms minority class detection, while retaining too few eliminates critical behavioral data. Furthermore, because these optimal feature subsets are highly tailored to the specific traffic profile of the training dataset, applying an optimized model directly to a different network environment without retraining would result in sub-optimal performance.

5.2. Performance evaluation and comparative analysis

After finding the best model for each dataset in the training and validation phase, the models were tested on the final test set. The primary evaluation compared the performance of the proposed models against established baseline models and published state-of-the-art results across

all three datasets. In the following comparisons, performance metrics are reported to four decimal places to maintain consistency with state-of-the-art literature and to distinguish between similar results. However, it is crucial to note that the primary objective of the OPF is not necessarily to achieve a massive raw increase in classification scores but rather to maintain these highly competitive, statistically similar performance levels while drastically reducing the feature space.

5.2.1. Comparison with established baseline models

Firstly, a comparison against established baseline models is performed. In fact, we compare our approach with: the baseline model, meaning no pre-processing technique is applied (neither dataset balancing nor feature selection), the Random Undersampled (RUS) model, which stands for a model where Random Undersampling is applied to the majority classes, and the SMOTE Oversampled model, where SMOTE Oversampling is applied to the minority classes. The results, presented in Tables 7, 8, and 9, consistently demonstrate the robustness of the OPF-based approach while using a reduced number of network features.

The results on the CICIoMT2024 dataset highlight the OPF's ability to optimize for the most critical performance metric in imbalanced classification. While the baseline model outperformed our approach in accuracy and precision, our model achieved a superior F1-score (0.9784). The higher F1-score confirms that our model provides the best trade-off between precision and recall, which is the primary goal of an effective IDS. This was accomplished while reducing the feature space by 31.11% and with a reduction in testing time. Moreover, the

Table 7

Performance and efficiency comparison on the CICIoMT2024 dataset.

Model	Validation F1-score	Test accuracy	Test precision	Test recall	Test F1-score	Features used (k)	Test time per sample (μs)
Baseline	0.9776	0.9989	0.9907	0.9658	0.9774	45	11.59
RUS	0.9474	0.9953	0.9340	0.9768	0.9491	45	11.58
SMOTE	0.9762	0.9988	0.9764	0.9756	0.9759	45	11.25
OPF (proposed)	0.9789	0.9988	0.9852	0.9721	0.9784	31	11.24

Table 8

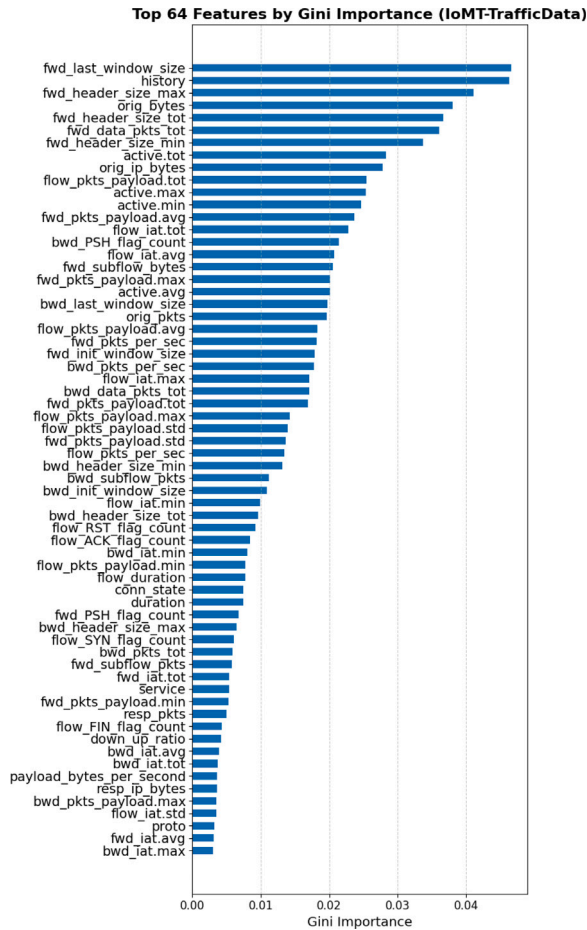
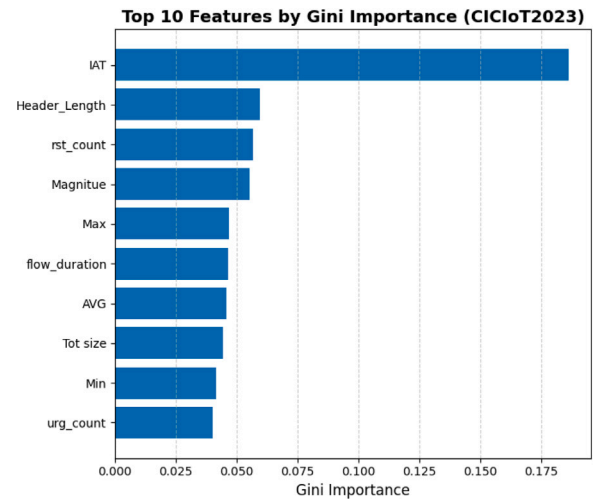
Performance and efficiency comparison on the IoMT-TrafficData dataset.

Model	Validation F1-score	Test accuracy	Test precision	Test recall	Test F1-score	Features used (k)	Test time per sample (μs)
Baseline	0.9979	0.9969	0.9476	0.9927	0.9629	94	30.18
RUS	0.9912	0.9958	0.9388	0.9937	0.9575	94	28.29
SMOTE	0.9980	0.9959	0.9442	0.9943	0.9605	94	27.08
OPF (proposed)	0.9981	0.9907	0.9365	0.9921	0.9524	64	30.16

Table 9

Performance and efficiency comparison on the CICIoT2023 dataset.

Model	Validation F1-score	Test accuracy	Test precision	Test recall	Test F1-score	Features used (k)	Test time per sample (μs)
Baseline	0.7904	0.9946	0.9829	0.7446	0.7669	46	21.77
RUS	0.7747	0.9776	0.8187	0.7689	0.7789	46	20.07
SMOTE	0.8473	0.9946	0.9829	0.7446	0.7969	46	20.80
OPF (proposed)	0.8684	0.9936	0.9598	0.8169	0.8707	10	20.03

**Fig. 6.** Top 64 features by Gini importance used in the IoMT-TrafficData dataset.**Fig. 7.** Top 10 features by Gini importance used in the CICIoT2023 dataset.

model also achieved high stability in the validation phase, with a higher average F1-score.

The IoMT-TrafficData results underscore the robustness and efficiency gains of the proposed framework. During the 5-fold cross-validation phase, the OPF model outperformed the baseline, RUS, and SMOTE models, achieving higher accuracy, recall, and F1-score, indicating superior model stability. On the final test set, while the baseline model achieved marginally higher scores, our OPF model's performance remained highly competitive and statistically similar. This slight variance can be attributed to the unique characteristics of the IoMT-TrafficData dataset, where we only take into consideration flow-based features. This suggests that an aggressive feature reduction might be slightly detrimental to the raw test F1-score when compared directly to the baseline model. Nevertheless, this pattern is a strong indicator of the baseline's tendency to overfit its 94 features, a risk that our method mitigates through its systematic feature space reduction. Critically, our model's robust performance was achieved with a 31.91% feature decrease (from 94 to 64 features), making it the far more

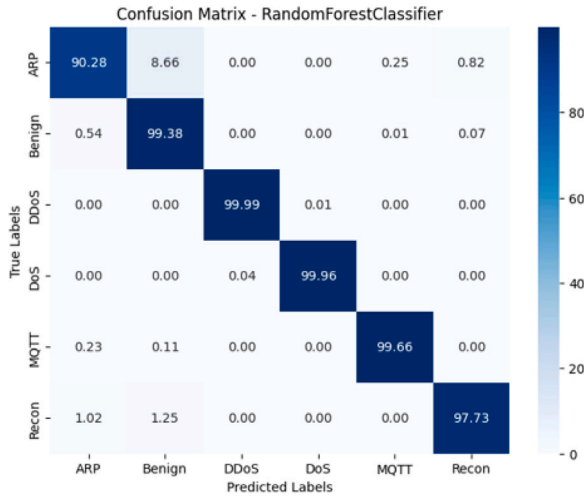


Fig. 8. Confusion matrix of the Baseline approach in the CICIoMT2024 dataset.

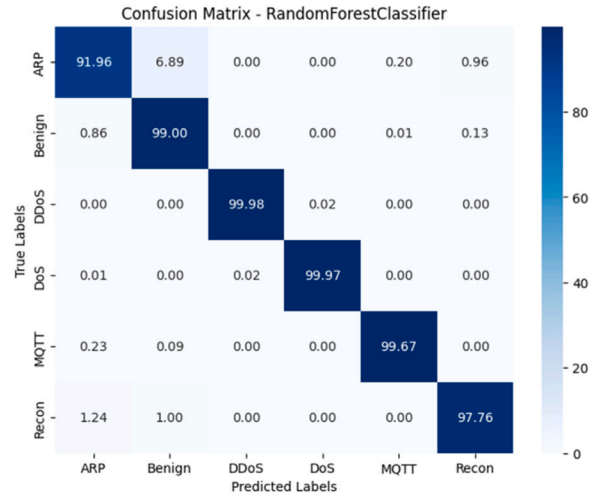


Fig. 9. Confusion matrix of the OPF approach in the CICIoMT2024 dataset.

practical, computationally efficient, and reliable choice for real-world deployment.

On the CICIoT2023 dataset, our OPF framework demonstrated clear superiority. The model not only achieved the highest F1-score (0.8707) and recall (0.8169) on the test set, but also showed greater stability with a higher average F1-score during cross-validation. While the baseline and the SMOTE model registered a marginally higher accuracy and precision, our model's superior recall and F1-score indicate a more effective overall balance for identifying threats. This high performance was achieved using 78.16% fewer features and with a reduction in testing time. This high variance and substantial improvement can be directly attributed to the extreme intra-class variance present in this dataset. While the models are trained to classify a standard number of high-level categories (7 attack classes plus benign traffic), the CICIoT2023 dataset is uniquely composed of 33 highly diverse sub-attack types grouped into these categories. Grouping distinct attack geometries under single labels can introduce feature noise, causing the baseline model to struggle with conflicting data distributions. By isolating only the 10 most critical features, the OPF is able to filter out the noise from these sub-attacks. This forces the model to learn only the fundamental attributes of the broader categories, resulting in a much larger performance compared to the other datasets.

We can also observe the major model's robust detection capabilities across heavily imbalanced minority classes in the confusion matrices with a better detection rate. For instance, Figs. 8 and 9 illustrate the confusion matrix of the baseline and the OPF approaches, respectively. An increase in detection rate is observed in the ARP Spoofing (1.68%), the DoS (0.01%), the MQTT (0.01%), and the Reconnaissance (0.03%) classes.

Figs. 10 and 11 present the confusion matrices of the baseline and OPF approach for the IoM-TrafficData. Therefore, an increase in the detection rate is observed in some classes such as Apache Killer (0.01%), ARP Spoofing (0.4%), R-U-Dead-Yet (0.23%), and Slow Read (1.5%).

Finally, we also improved the detection rate of some classes in the CICIoT2023 dataset. Specifically, a major improvement of 30.21% and 27.49% is seen in the BruteForce and the Web classes, respectively. Other classes, such as Reconnaissance (3.18%) and Spoofing (0.5%), experienced an improvement in the detection rate, as illustrated in Figs. 12 and 13.

5.2.2. Comparison with state-of-the-art models

Secondly, we compare our OPF model with existing state-of-the-art models. As observed in Table 10, our approach evaluated in the test set

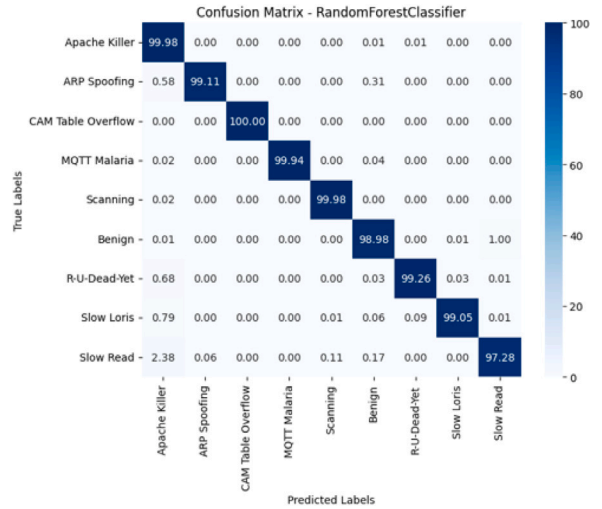


Fig. 10. Confusion matrix of the Baseline approach in the IoMT-TrafficData dataset.

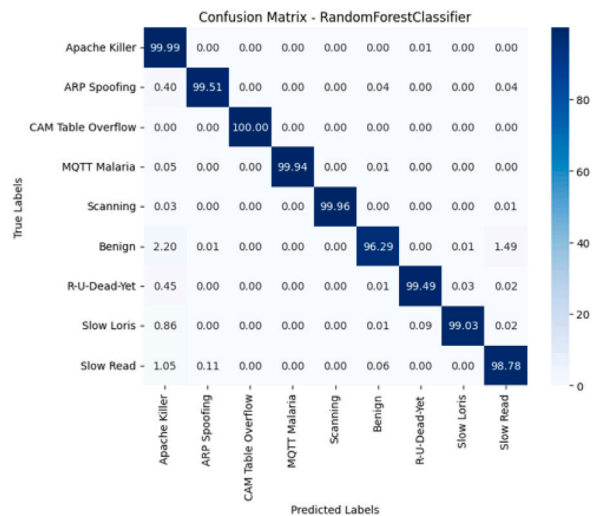


Fig. 11. Confusion matrix of the OPF approach in the IoMT-TrafficData dataset.

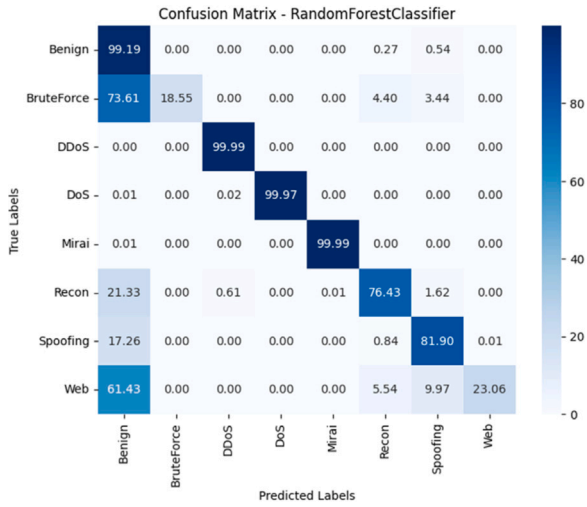


Fig. 12. Confusion matrix of the Baseline approach in the CICIoT2023 dataset.

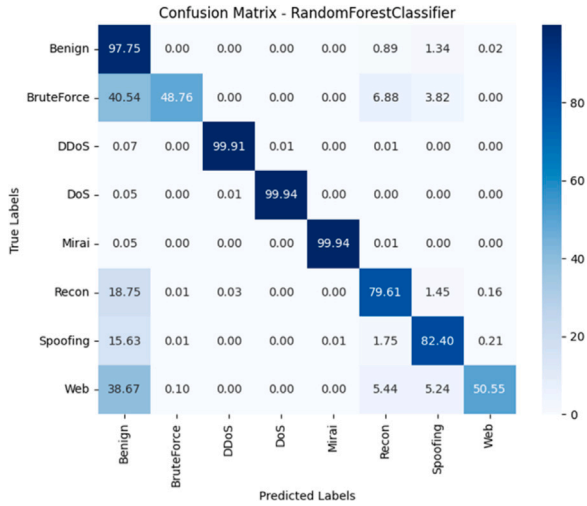


Fig. 13. Confusion matrix of the OPF approach in the CICIoT2023 dataset.

outperforms previous models on the CICIoMT2024 dataset in terms of accuracy, precision, recall, F1-score, and it reduces the features used by 31.11%. Moreover, the proposed model also has a higher F1-score (the most critical metric for imbalanced security datasets) than state-of-the-art models when testing it within the CICIoT2023 dataset, while reducing by 78.16% the feature space. This highlights our model's more effective balance between precision and recall. However, the OPF-test model does not have a higher performance than state-of-the-art models when using the IoMT-TrafficData, but still reduces the feature space by 31.91% while remaining highly competitive and statistically similar to the others.

A direct comparison of the OPF framework with existing state-of-the-art models reveals both the competitiveness of our approach and a significant challenge in the field: the lack of standardized evaluation protocols. For instance, a notable discrepancy can be observed in the CICIoMT2024 dataset: while our simple baseline model achieves an F1-score of 0.9774, recent state-of-the-art models such as the one proposed by Kouekam and Khennou [20] report a much lower F1-score of 0.7881. This vast difference does not imply that state-of-the-art algorithms are inherently flawed; rather, it highlights that prior work employs fundamentally different experimental configurations. Specifically, Kouekam and Khennou [20] utilize a different train-test data split (75-25 instead of 80-20) and does not specify whether a validation split was

used, nor whether the test split was completely unseen by the model. Because different architectures, data scaling procedures, and test-set splits heavily skew comparative metrics. For that reason, we explicitly included all the steps of our methodology, which guarantees a fair evaluation. Furthermore, to facilitate a transparent and multi-faceted comparison, in Table 10 we include both our final model evaluated on the test set (OPF-test) and our average 5-fold cross-validation results (OPF-validation).

The comparison of the three datasets underscores the critical importance of methodological transparency and the necessity of a pre-agreed protocol for fair model performance comparison. Specifically, in the IoMT-TrafficData analysis, while our final test set F1-score is competitive yet lower than the reported by Areia et al. [44] and Adeyemi et al. [14], our average cross-validation F1-score of 0.9981 overcomes their reported results. This discrepancy suggests that our model is more statistically robust and that some state-of-the-art models may reflect performance on validation data rather than truly unseen test data. Given this ambiguity, our cross-validation performance may represent a more equitable comparison point. Critically, across all three datasets, the OPF consistently requires a significantly smaller feature set than any other model, confirming its superior data efficiency and suitability for practical deployment.

5.3. Limitations and challenges

While our proposed OPF framework demonstrates significant advantages in building data-efficient and robust AI-based IDS, reducing sensitivity against class imbalance; it is important to acknowledge the limitations of this study, which can give promising directions for future research.

A key observation of our results is that a single model rarely dominates across all performance metrics simultaneously. This reflects a specific trade-off in machine learning for imbalanced classification. The choice of which metric to prioritize depends on the specific security policy of an organization, and our framework, designed for medical environments, is primarily tuned for the balanced F1-score to have a good stability between precision and recall.

Moreover, as discussed in Section 5.2, our comparative analysis highlighted a significant challenge in the field: the lack of standardized evaluation protocols. Many studies do not explicitly detail their data splitting methodologies, making it difficult to ascertain if reported results are from a truly unseen test set [10]. Our work tries to mitigate this challenge by transparently reporting both validation and test results.

Our approach is only trained, validated, and tested using three well-known datasets: CICIoMT2024, IoMT-TrafficData, and CICIoT2023. We know that the OPF is a robust and data-efficient solution in these environments, but further research should be done by evaluating our model on other publicly available IoT/IoMT datasets or real environments. Moreover, the evaluated datasets are generated in testbeds, which may not capture the full complexity of live hospital networks. However, there are no real hospital network datasets publicly available in the scientific literature, making this analysis limited to testbed datasets.

Finally, it is important to clarify the scope of *robustness* achieved by the OPF. While our framework demonstrates significant robustness against extreme class imbalance and overfitting, evaluating its resilience against adversarial evasion attacks, data poisoning, or injected network noise was outside the scope of this study.

6. Conclusion and future work

This study addressed the critical challenge of designing a data-efficient Intrusion Detection System that is robust to the highly imbalanced nature of IoMT network traffic by proposing the Optimized Pre-processing Framework (OPF). This novel two-stage approach integrates feature selection and data balancing to solve the problem

Table 10
State-of-the-art comparison on the CICIoMT2024, IoMT-TrafficData, and CICIoT2023 datasets.

Dataset	Model	Accuracy	Precision	Recall	F1-score	Features used (<i>k</i>)
CICIoMT2024	Dadkhah et al. [28]	0.7350	0.7130	0.7650	0.6760	45
	Kouekam and Khennou [20]	0.7928	0.7928	0.7924	0.7881	45
	OPF-validation (proposed)	0.9988	0.9858	0.9723	0.9789	31
	OPF-test (proposed)	0.9988	0.9852	0.9721	0.9784	31
IoMT-TrafficData	Areia et al. [44]	0.9952	0.9952	0.9952	0.9952	94
	Gül and Bakir [21]	0.9957	0.9957	0.9957	0.9957	94
	OPF-validation (proposed)	0.9996	0.9984	0.9979	0.9981	64
	OPF-test (proposed)	0.9907	0.9365	0.9921	0.9524	64
CICIoT2023	Neto et al. [45]	0.9943	0.7054	0.9100	0.7192	46
	Adeyemi et al. [14]	0.9970	0.9858	0.7539	0.8160	46
	OPF-validation (proposed)	0.9936	0.9587	0.8146	0.8684	10
	OPF-test (proposed)	0.9936	0.9598	0.8169	0.8707	10

of high-dimensionality and the imbalanced nature of IoMT network data. The framework's key contributions include the use of an efficient Gini feature-based prioritization stage and a subsequent hybrid resampling stage that combines SMOTE-based oversampling with random undersampling to create a nearly balanced dataset, thereby avoiding overfitting and mitigating majority class dominance while preserving representative diversity.

A comprehensive evaluation of three recent and well-known datasets demonstrated that OPF can be used in different environments. Moreover, the resulting model achieves state-of-the-art detection performance, significantly outperforming the baseline and published models, especially in terms of F1-score, with validation scores of 0.9789, 0.9981, and 0.8684, and test scores of 0.9784, 0.9524, and 0.8707 in the CICIoMT2024, IoMT-TrafficData, and CICIoT2023 datasets, respectively. This high performance was achieved alongside a considerable reduction in feature space (up to 78.16%) and a corresponding decrease in computational overhead. These findings confirm that the OPF presents a practical and effective solution for securing real-world IoMT ecosystems.

This work suggests several directions for future research that can build upon the foundation of the OPF. While this study focuses on three well-known datasets, future work should explore the integration of the framework with other datasets for attack detection, both in IT and IoT domains. Specifically, there is a critical need to transition from academic testbeds to real-world hospital network data to validate the framework's adaptability to real environments. Additionally, future research must address the operational security of the IDS itself by evaluating its resilience against adversarial evasion attacks. This must include an interdisciplinary analysis of IDS failure modes to understand the specific medical consequences and patient safety risks associated with false negatives in a clinical environment. Furthermore, the current framework is trained, validated, and tested in specific datasets; a significant and challenging direction for future work would be to adapt the OPF and the final feature space to be able to generalize to other environments, selecting features that are not dependent on the environment where the model was trained. Moreover, the current empirical grid search for optimal resampling ratios is effective but computationally intensive; future research could focus on automating and optimizing this process. Finally, the OPF is designed and tested in offline scenarios; further work should focus on the deployment of the model in real-time. These advancements would contribute to developing more reliable and scalable IDS solutions, ensuring the security and safety of IoMT environments.

CRediT authorship contribution statement

Jordi Doménech: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Saber Mhiri:** Writing – review & editing, Writing

– original draft, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Olga León:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Muhammad Shuaib Siddiqui:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Formal analysis. **Josep Pegueroles:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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